

JEE MAIN REVISION MASTER: FUNCTIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 158 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Functions - Core Application	158 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Functions - Core Application (Very High Yield — 158 PYQs)

Core Principle: Key problem pattern in Functions.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $f:R \rightarrow R$ be defined as $f(x)=\frac{2x^2-3x+23x^2+x+3}{x^2+1}$. Then f is :..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let $[\cdot]$ denote the greatest integer function. If the domain of the function $f(x)=\cos^{-1}((4x+2[x] / 3))$ is $[\alpha, \beta]$, then $\beta - \alpha =$..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "If $g(x)=3x^2+2x-3, f(0)=-3$ and $4g(f(x))=3x^2-32x+72$, then $f(g(2))$ is equal to:..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

Let $[\cdot]$ denote the greatest integer function. If the domain of the function $f(x) = \cos^{-1}((4x + 2[x] / 3))$ is $[\alpha, \beta]$, then $12(\alpha + \beta)$ is equal to :

- (A) 6 **[Correct]**
- (B) 8
- (C) 9
- (D) 4

Correct Answer: A

Solution Analysis: For the function $f(x) = \cos^{-1}((4x + 2[x] / 3))$, the expression inside $\cos^{-1}$ must lie between -1 and 1. So, for domain: $-1 \leq (4x + 2[x] / 3) \leq 1$ Multiplying by 3 : $-3 \leq 4x + 2[x] \leq 3$ Now let $[x] = n$ where n is an integer and $n \leq x < n+1$. Then the inequality becomes $-3 \leq 4x + 2n \leq 3$. Subtracting $2n$: $-3 - 2n \leq 4x \leq 3 - 2n$ Dividing by 4 : $(-3 - 2n / 4) \leq x \leq (3 - 2n / 4)$ But also, $n \leq x < n+1$. So we find possible integer values of n for which these intervals overlap. Case 1: $n = -1$ Then $[x] = -1, -1 \leq x < 0$ and $\frac{-3 - 2(-1)}{4} \leq x \leq \frac{3 - 2(-1)}{4}$